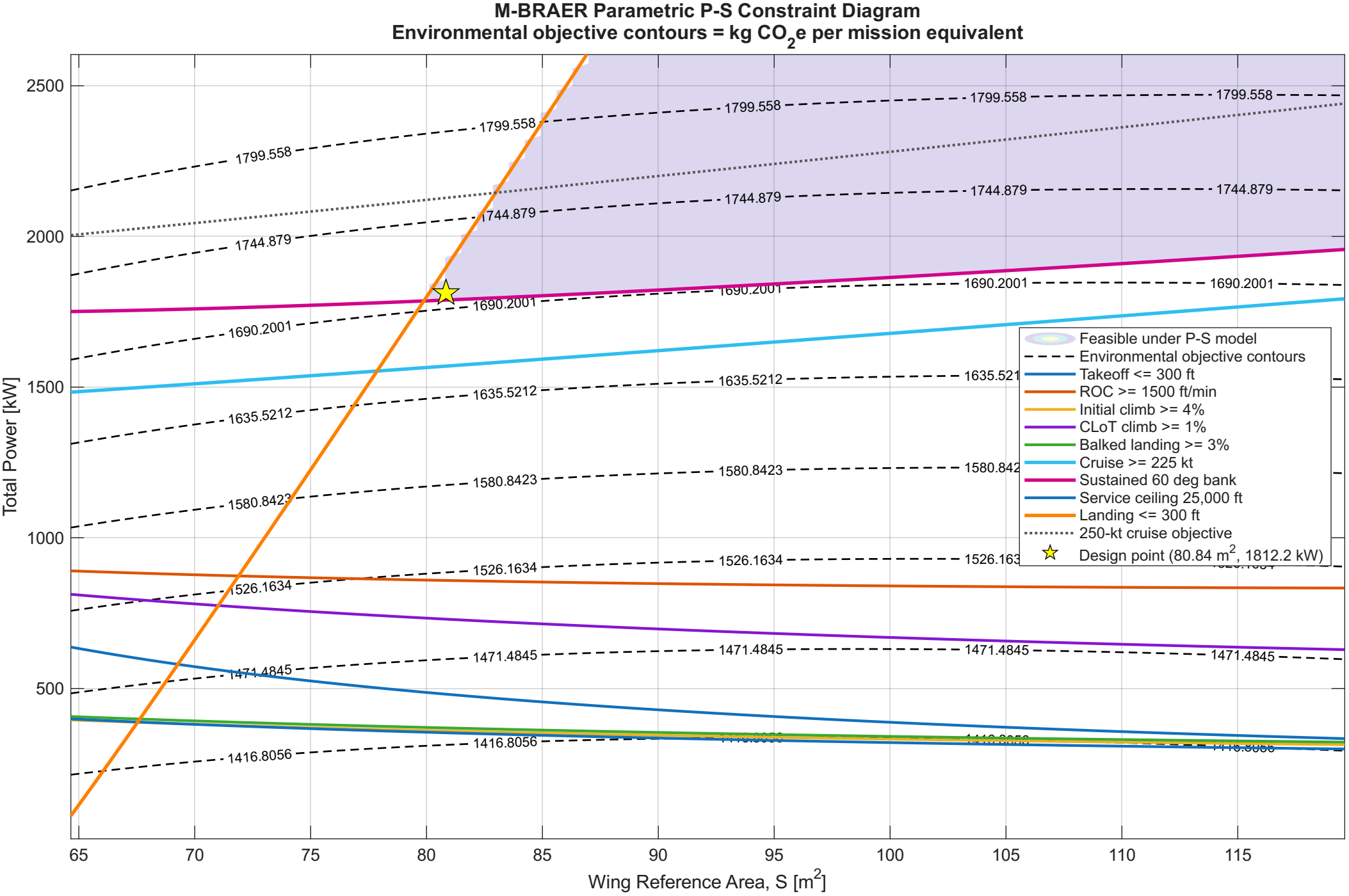
M-BRAER Parametric P-S Constraint Diagram
Environmental objective contours = kg CO2e per mission equivalent
Total Power [kW]
Wing Reference Area, S [m2]
2500
2000
1500
1000
500
65
70
75
80
85
90
95
100
105
110
115
1799.558
1744.879
1690.2001
1635.5212
1580.8423
1526.1634
1471.4845
1416.8056
Feasible under P-S model
Environmental objective contours
Takeoff <= 300 ft
ROC >= 1500 ft/min
Initial climb >= 4%
CLoT climb >= 1%
Balked landing >= 3%
Cruise >= 225 kt
Sustained 60 deg bank
Service ceiling 25,000 ft
Landing <= 300 ft
250-kt cruise objective
Design point (80.84 m2, 1812.2 kW)